GSSS INSTITUTE OF ENGINEERING AND TECHNOLOGY FOR WOMEN,MYSURU
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING) MLOps ASSIGNMENT – 2

Tracking a Pre-trained Sentiment Model with MLflow

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1. AIM

To use a pre-trained sentiment analysis model and track its inference experiments using MLflow. The experiments are performed with different batch sizes (1, 8, and 4), and their parameters, metrics, and prediction results are recorded and compared.

2. SOFTWARE REQUIREMENTS

- **Operating System:** Windows
- **Programming Language:** Python 3.12.10
- **IDE:** Visual Studio Code
- **Libraries:** MLflow, Transformers, PyTorch, and Pandas
- **Pre-trained Model:** distilbert-base-uncased-finetuned-sst-2-english

3. INTRODUCTION

MLOps (Machine Learning Operations) is a set of practices used to manage machine learning models and their workflows. It helps organize experiments, record results, and make model development and deployment easier to manage.

MLflow is an open-source platform used to track machine learning experiments. It records parameters (settings used in an experiment), metrics (measured results), and artifacts (files produced during an experiment). Its user interface allows different experiment runs to be viewed and compared.

In this assignment, a pre-trained DistilBERT sentiment analysis model is used. The model classifies sentences as POSITIVE or NEGATIVE. No model training is performed; only inference is carried out on ten labelled sample sentences.

4. PROCEDURE

1. Create a project folder and open it in Visual Studio Code.
2. Create and activate a Python 3.12 virtual environment.
3. Install MLflow, Transformers, PyTorch, and Pandas.
4. Load the pre-trained DistilBERT sentiment analysis model.

5. Prepare ten sentences with their expected sentiment labels.
6. Run inference using batch sizes 1, 8, and 4.
7. For each run, log the model name, batch size, and number of samples as parameters.
8. Calculate and log accuracy, average confidence, and total inference latency as metrics.
9. Save the input sentences, expected labels, predicted labels, and confidence scores in CSV files.
10. Open the MLflow user interface and compare the experiment runs.

5. EXPERIMENTAL RESULTS

The following results were obtained during the successful execution of the program. Latency may vary slightly between executions depending on system load and runtime conditions.

Run	Batch Size	Accuracy	Average Confidence	Latency (seconds)
run_batch1	1	1.00 (100%)	0.9998	0.2537
run_batch8	8	1.00 (100%)	0.9998	0.1131
run_batch4	4	1.00 (100%)	0.9998	0.1181

Observation: All three runs achieved 100% accuracy on the ten sample sentences. Batch size 8 achieved the lowest observed latency of approximately 0.1131 seconds.

Note: Accuracy is calculated against the expected labels supplied for these ten sample sentences. It does not guarantee the same accuracy on other datasets.

6. QUESTIONS AND ANSWERS

1. Which run had the lowest latency, and why might that be?

The run_batch8 experiment had the lowest observed latency of approximately 0.1131 seconds, followed by run_batch4 at 0.1181 seconds and run_batch1 at 0.2537 seconds.

A larger batch size allows the model to process multiple sentences together instead of processing them one by one. This can reduce processing overhead and improve inference speed.

2. Did accuracy change between runs? Why or why not?

No, the accuracy remained the same at 1.00 (100%) in all three runs.

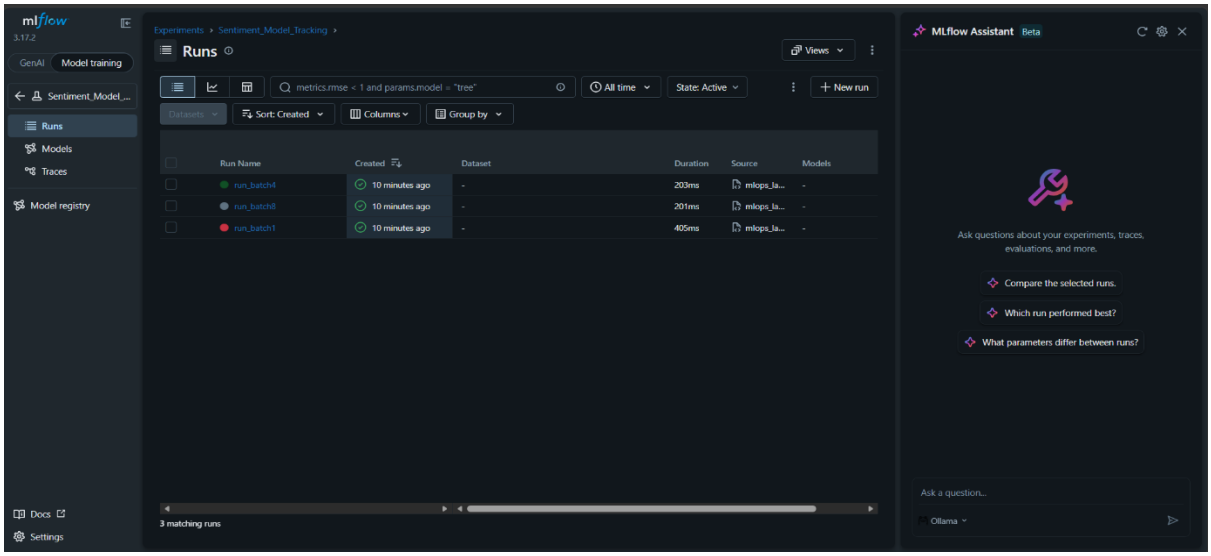
Each run used the same pre-trained sentiment analysis model and the same ten sample sentences. Only the batch size was changed. Therefore, the predicted sentiment labels remained the same in this execution, and the accuracy did not change.

3. Name two benefits of using MLflow here instead of printing results in the terminal.

1. Reproducibility and record-keeping: MLflow automatically stores experiment parameters, metrics, and artifacts. This helps us review previous experiment results whenever needed.

2. Easy comparison: MLflow provides a user-friendly dashboard to compare different experiment runs side by side. We can easily compare accuracy, latency, and other metrics using tables and visualizations.





7. CONCLUSION

The pre-trained DistilBERT sentiment analysis model was successfully used to perform sentiment inference with batch sizes 1, 8, and 4.

MLflow recorded the experiment parameters, metrics, and prediction CSV artifacts. The experiment runs were viewed and compared using the MLflow user interface.

In this execution, batch size 8 achieved the lowest observed inference latency while maintaining 100% accuracy on the supplied sample dataset. This experiment demonstrates how MLflow helps track, organize, and compare machine learning experiments efficiently.